

Set :-

- 1) Set is also one of the container data types in python.
- 2) Set is similar to the list, but exception is that duplicate elements are not allowed in sets.
- 3) list $\rightarrow$ [10, 20, 30, 40, 50]
tuple $\rightarrow$ (10, 20, 30, 40, 50)
set $\rightarrow$ {10, 20, 30, 40, 50}

Set is defined using curly brackets & elements are separated by comma.

- 4) We can have homogeneous elements in the set

set1 = {"ganesha", "mahesh", "rajesha"}

5) We can have heterogeneous elements in the set

Set 2 = $\{10, 10.1, \text{True}, \text{"gonesh"}, 2+3j\}$

6) Duplicate elements are not allowed in set

eg. Set 3 = $\{10, 20, 30, 40, 50, 10\}$

print(Set3)

o/p- $\{10, 20, 30, 40, 50\}$

7) When we store element in the set it calculates hash value of every element using hashing technique.

& using this hash value it decides if particular element will be stored in the set or not.

Set 4 = $\{10, 20, 30, 40, 50, 10\}$

10 - 1000

20 - 1001

30 - 1002

40 - 1003

50 - 1004

10 - 1000X

o/p -

{50, 30, 20, 10, 40}

To calculate hash value

print(hash(10.1)) → --

print(hash("gresh")) → --

8) we can create empty set

eg. set6 = set()

print(set6) → {}

print(type(set6)) → <set>

empty list → list1 = []
list2 = list()

empty tuple → tuple1 = ()
tuple2 = tuple()

empty set → set1 = set() ✓
X

empty dict $\rightarrow$ dict1 = {} $\rightarrow$ dict
dict2 = dict()

9) Set is Unordered. It does not preserve the sequence of insertion

eg. set1 = {10, 20, 30, 40, 50}

print(set1)

o/p - {50, 30, 40, 10, 20}

list, tuple, string, dict - Ordered
set - Unordered

list1 = [10, 20, 30, 40, 50]

print(list1) $\rightarrow$ [10, 20, 30, 40, 50]

name = "gareesh"

print(name) $\rightarrow$ "gareesh"

- Accessing set elements

We can access entire set using
set name.

eg. Set1 = {10, 20, 30, 40, 50}

print(Set1)

o/p - {50, 20, 10, 40, 30}

Set is an Unordered Collection,
& hence we cannot use
indexing & slicing.

Set1 = {⁰10, ¹20, ²30, ³40, ⁴50}

looping in set :-

1] for loop -

Set1 = {10, 20, 30, 40, 50}

for i in Set1:

print(i)

o/p -
50
30
10
40
20

Basic set operations -

1] Mutability - Set is mutable.

- ① String - immutable
- ② List - mutable
- ③ Tuple - immutable
- ④ Set - mutable
- ⑤ dict - mutable

Basic data types

int, float, complex, bool → immutable

eg. Set 1 = { 10, 20, 30 }

Set 1.add(40)

print (Set 1)

o/p - { 10, 20, 30, 40 }

add
delete
replace

2] Conversion - we can convert

string, list, tuple, dict, range()
into set.

Built-in function - set()

i) Converting string to set -

```
str1 = "hello"
```

↓

```
set1 = set(str1)
```

```
print(set1) → {'h', 'e', 'l', 'o'}
```

ii) Converting list into set

```
list1 = [10, 20, 30]
```

↓

```
set2 = set(list1)
```

```
print(set2) → {10, 20, 30}
```

iii) Converting tuple into set

```
tuple1 = (10, 20, 30)
```

↓

```
set3 = set(tuple1)
```

```
print(set3) → {10, 20, 30}
```

iv) Converting dict into set

```
dict1 = { "name" : "ganesh",  
          "age" : 25 }
```

age : 25 }

set4 = set(dict1)

print(set4) → { 'name', 'age' }

v) range() into set

1, 2, 3 ... 10

range1 = range(1, 11, 1)

set5 = set(range1)

print(set5) → { 1, 2, 3 ... 10 }

3. Aliasing -

Set1 = { 10, 20, 30 }

set2 = set1

print(id(set1))

print(id(set2))

→ Same

set2.add(40)

print(set1) → { 10, 20, 30, 40 }

4. Cloning -

Set1 = { 10, 20, 30 }

```
set2 = set1.copy()
```

```
print(id(set1))
```

```
print(id(set2))
```

} different

```
set2.add(40)
```

```
print(set1) → {10, 20, 30}
```

```
print(set2) → {10, 20, 30, 40}
```

5) identity - 'is' operator

```
set1 = {10, 20, 30}
```

```
set2 = set1
```

```
print(set1 is set2) → True
```

6. Searching - 'in' operator

```
set1 = {10, 20, 30}
```

```
print(10 in set1) → True
```

```
print(20 in set1) → True
```

print(100 in set1) → False

print("ganesb" in set1) → False

7. Emptiness - not operator

set1 = set()

print(not set1) → True

set2 = {10, 20, 30}

print(not set2) → False

Built in functions on set -

1] len() - returns length of set

eg. set1 = {10, 20, 30}

print(len(set1)) → 3

2] max() - returns maximum element

set1 = { 10, 20, 30 }

print (max (set1)) → 30

3] min() - returns minimum element

set1 = { 10, 20, 30 }

print (min (set1)) → 10

4] sum() - returns addition of all elements.

set1 = { 10, 20, 30 }

print (sum (set1)) → 60

5] any() - it returns true if any element is true of set

set1 = { 0, 0, 0, 1 }

print (any (set1)) → True

6] all() - it returns true if all elements are true in the set

Set1 = { 10, 20, 30, 40 }

print (all (Set1)) → True

Set2 = { 10, 20, 30, 0 }

print (all (Set2)) → False

7] Sorted() - it returns sorted list

Set1 = { 10, 50, 40, 20, 30 }

print (sorted (Set1))

o/p - [10, 20, 30, 40, 50]